ANALYSTLAB AFRICA

DATA ANALYTICS INTERNSHIP

WEEK 5 PROJECT REPORT

SwiftLog Logistics Analytics & AI Transformation Project

Title: Enterprise supply chain Analytics and Artificial Intelligence Transformation for SwiftLog Logistics Limited.

TOOLS: Microsoft Power BI Desktop (Power Querry and DAX)

INTERN: OTUTU LINDA CHINECHEREM

SwiftLog Logistics Analytics & AI Transformation Project

Part 2: Data Cleaning & Transformation

1. Introduction

This document records every transformation applied to the 11 SwiftLog datasets in Power Query. Each action is stored as a named step in the Applied Steps panel of its query, which forms an auditable record. Column names from the source were kept unchanged so that keys match across tables and the DAX measures continue to work. New columns were added with clear descriptive names.

2. Standard Cleaning Routine

The following routine was applied to every table:

- Checked the first row was used as headers (Use First Row as Headers).
- Set each column to its correct data type: IDs as Text, OrderDate as Date, Quantity, DelayDays and Stock as Whole Number, UnitPrice as Fixed Decimal Number.
- Trimmed and cleaned all text columns (Transform, Format, Trim, then Clean) and capitalised descriptive text (Capitalize Each Word).
- Checked for errors and empty values using Column quality. None were found, so no Remove Errors, Replace Errors or Remove Empty steps were needed.
- Checked key columns for duplicates. Only Inventory had exact duplicate rows.

3. Transformations by Tables

3.1 SwiftLog_Customers (1,000 rows)

No.	Transformation	How it was done in Power Query	Result
1	Verified data types	Set all three columns to Text	Correct types
2	Standardised text	Trim, Clean, Capitalize Each Word on CustomerName and City	Consistent text
3	Split a column	Duplicated CustomerName; Transform, Split Column, By Delimiter (space, left-most)	Customer Prefix and Customer Number columns
4	Customer Number type	Changed Customer Number to Whole Number	Numeric field

3.2 SwiftLog_Orders (5,000 rows)

No.	Transformation	How it was done in Power Query	Result
1	Converted dates	Right-click OrderDate, Change Type, Using Locale, Date, English (United States)	Dates read correctly as month/day/year

No.	Transformation	How it was done in Power Query	Result
2	Standardised text	Trim, Clean on OrderID, CustomerID, Status; Capitalize Each Word on Status	Consistent status values
3	Date columns	Add Column, Date: Year, Month, Name of Month, Quarter of Year, Day	Order Year, Order Month Number, Order Month Name, Order Quarter Number, Order Day
4	Quarter label	`Custom column Order Quarter: "Q" & Text.From([Order Quarter Number])`	Q1 to Q4 labels
5	Custom business fields	`Is Delivered: if [Status] = "Delivered" then 1 else 0 Is Cancelled: if [Status] = "Cancelled" then 1 else 0`	Flags used for KPIs

3.3 SwiftLog_Products (250 rows)

No.	Transformation	How it was done in Power Query	Result
1	Set types	ProductID and Category as Text; UnitPrice as Fixed Decimal Number	Correct types
2	Standardised text	Trim, Clean, Capitalize Each Word on Category	Consistent categories
3	Conditional column	`Price Tier (Add Column, Conditional Column): UnitPrice < 100 then "Budget" else if < 300 then "Standard" otherwise "Premium"`	Price Tier

3.4 SwiftLog_Order_Items (15,000 rows)

No.	Transformation	How it was done in Power Query	Result
1	Set types	IDs as Text; Quantity as Whole Number	Correct types
2	Merge tables	Home, Merge Queries: Order_Items and Products on ProductID, Left Outer; expanded UnitPrice only	UnitPrice added; row count stays 15,000
3	Calculated column	`Line Revenue (Custom Column): [Quantity] * [UnitPrice]`	Revenue per order line
4	Conditional column	`Order Size: Quantity >= 8 then "Bulk" else if >= 4 then "Standard" otherwise "Small"`	Order Size

3.5 SwiftLog_Shipments (5,000 rows)

No.	Transformation	How it was done in Power Query	Result
1	Set types	IDs as Text; DelayDays as Whole Number	Correct types

2	Conditional column	`Delay Category:`
		`DelayDays = 0 then "No Delay"`
		`else if <= 2 then "Within Tolerance"`
		`else if <= 5 then "Moderate Delay"`
		`otherwise "Severe Delay"`
			Delay Category

3	Custom business field	`Is Delayed:` `if [DelayDays] > 2 then 1 else 0`	Delay flag

3.6 SwiftLog_Delivery_Performance (5,000 rows)

No.	Transformation	How it was done in Power Query	Result
1	Set types	ShipmentID and Status as Text	Correct types
2	Removed an unnecessary column	Removed DelayDays, which is identical to Shipments[DelayDays] (0 mismatches found)	Redundant data removed
3	Standardised text	Trim, Clean, Capitalize Each Word on Status	On Time and Delayed

3.7 SwiftLog_Returns (500 rows)

No.	Transformation	How it was done in Power Query	Result
1	Set types	All columns as Text	Correct types
2	Standardised text	Trim, Clean, Capitalize Each Word on Reason	Consistent reasons

3.8 SwiftLog_Warehouses (15 rows)

No.	Transformation	How it was done in Power Query	Result
1	Set types and standardise	Text types; Trim, Clean, Capitalize Each Word on Location	Consistent locations
2	Merge columns	Add Column, Merge Columns on WarehouseID and Location, separator " - " (Add Column tab keeps the originals)	Warehouse Label, for example W01 - Abuja

3.9 SwiftLog_Employees (200 rows)

No.	Transformation	How it was done in Power Query	Result
1	Set types and standardise	Text types; Trim and Clean	Correct types

3.10 SwiftLog_inventory (4,000 rows reduced to 3,997)

No.	Transformation	How it was done in Power Query	Result
1	Set types	IDs as Text; Stock as Whole Number	Correct types
2	Removed duplicates	Selected WarehouseID, ProductID and Stock together; Remove Duplicates	3 exact duplicate rows removed

3	Conditional column	`Stock Level:`
		`Stock < 100 then "Low"`
		`else if < 300 then "Medium"`
		`otherwise "High"`
			Stock Level

4	Surrogate key	Add Column, Index Column, From 1, named Inventory ID	Unique key for the table

3.11 SwiftLog_Suppliers (150 rows)

No.	Transformation	How it was done in Power Query	Result
1	Set types and standardise	Text types; Trim, Clean, Capitalize Each Word on SupplierName	Correct types

4.0 Date Table

A date table was created with DAX (Modeling, New Table) and marked as the date table. It covers 1 January to 31 December 2026 and has Year, Month Number, Month Name and Quarter columns. It is related to SwiftLog_Orders[OrderDate] one-to-many and supports time-based analysis.

Item	Detail

Table	`Dim_Date =`
	`ADDCOLUMNS(CALENDAR(DATE(2026,1,1), DATE(2026,12,31)),`
	`"Year", YEAR([Date]),`
	`"Month Number", MONTH([Date]),`
	`"Month Name", FORMAT([Date], "MMMM"),`
	`"Quarter", "Q" & ROUNDUP(MONTH([Date])/3, 0))`

Settings	Table tools, Mark as date table (Date column); Month Name sorted by Month Number

5.0 Summary of Required Transformations

Requirement	Where applied
Clean all datasets	All 11 tables (standard routine, section 2)
Remove unnecessary columns	Delivery_Performance: DelayDays (duplicate of Shipments)
Standardise text values	Trim, Clean and Capitalize Each Word on all descriptive text columns
Split columns	Customers: CustomerName split into Customer Prefix and Customer Number
Merge columns	Warehouses: Warehouse Label

Requirement	Where applied
Merge tables	Order_Items and Products (UnitPrice)
Date, Month, Quarter, Year columns	Orders: Order Year, Order Month Number, Order Month Name, Order Quarter Number, Order Quarter, Order Day
Conditional columns	Price Tier, Order Size, Delay Category, Stock Level
Calculated columns	Line Revenue
Custom business fields	Is Delivered, Is Cancelled, Is Delayed, Warehouse Label, Inventory ID

6. Data Dictionary: Columns Added in Power Query

Column	Table	Logic	Type
Customer Prefix, Customer Number	Customers	Split of CustomerName at the space	Text, Whole Number
Order Year, Order Month Number, Order Day	Orders	Taken from OrderDate	Whole Number
Order Month Name	Orders	Month name from OrderDate	Text
Order Quarter Number, Order Quarter	Orders	Quarter 1 to 4, and the label Q1 to Q4	Whole Number, Text
Is Delivered, Is Cancelled	Orders	1 when Status matches, otherwise 0	Whole Number
Price Tier	Products	Budget under 100, Standard under 300, otherwise Premium	Text
UnitPrice (merged)	Order_Items	Looked up from Products by ProductID	Fixed Decimal
Line Revenue	Order_Items	Quantity multiplied by UnitPrice	Fixed Decimal
Order Size	Order_Items	Bulk 8 or more, Standard 4 to 7, Small under 4	Text
Delay Category	Shipments	No Delay, Within Tolerance, Moderate Delay, Severe Delay	Text
Is Delayed	Shipments	1 when DelayDays is more than 2, otherwise 0	Whole Number
Warehouse Label	Warehouses	WarehouseID and Location joined with " - "	Text
Stock Level	Inventory	Low under 100, Medium under 300, otherwise High	Text
Inventory ID	Inventory	Index starting at 1	Whole Number

7. Transformation Log

Dataset	Issue found	Fix applied
Inventory	3 exact duplicate rows	Remove Duplicates on all three columns (4,000 to 3,997 rows)
Delivery_Performance	DelayDays duplicates Shipments[DelayDays]	Removed the column from Delivery_Performance
Orders	OrderDate stored as text in M/D/YYYY format	Changed type to Date using the English (United States) locale
Order_Items	No price or revenue available	Merged UnitPrice from Products and created Line Revenue
All tables	No missing values, errors or spacing problems	Checked and documented; Trim and Clean applied as standard practice
All tables	Descriptive fields needed for analysis	Added date, conditional, calculated and custom columns (section 6)

8. Row Counts Before and After

Table	Before	After
Customers	1,000	1,000
Orders	5,000	5,000
Order_Items	15,000	15,000
Products	250	250
Shipments	5,000	5,000
Delivery_Performance	5,000	5,000
Returns	500	500
Warehouses	15	15
Employees	200	200
Inventory	4,000	3,997
Suppliers	150	150

Merging UnitPrice into Order_Items must not change its row count. The unchanged count of 15,000 confirms the merge did not duplicate rows.

9. Model Optimisation

- Correct data types on every column, with Whole Number used for counts and days.
- Removed the redundant DelayDays column from Delivery_Performance.
- Star-schema style model with fact tables in the centre and dimension tables around them.
- One-to-many relationships with single-direction filtering.

- A dedicated date table marked as the date table, and the automatic date/time option switched off (File, Options, Current File, Data Load).
- Key and foreign-key columns hidden from Report view so that report users see only analytical fields.
- All measures stored together, with consistent number formats.

10. Conclusion

All 11 datasets were cleaned and transformed in Power Query, and each action is documented and traceable through the Applied Steps. The result is a consistent, correctly typed and well-structured model with the new fields needed for the KPIs in Part 3 and the dashboard in Part 4.